Stability Control and Path Tracking of a Self-Balancing Bicycle With a Reaction Wheel

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***Abstract*—This article introduces a self-balancing bicycle utilizing a reaction wheel for stability control. To manage hardware complexity, a simplified test platform was first used for physical modeling and controller design with a linear-quadratic regulator (LQR), validating the concept. For the final implementation, the full-scale bicycle adopts full-state feedback. Although the LQR improved stability, its frequent motor speed adjustments led to high power consumption. To address this, a novel adaptive equilibrium point gradual algorithm (AEPGA) was developed to adapt to environmental changes, such as lateral imbalances. Performance tests on a real bicycle showed the AEPGA achieved maximum speed of 17 km/h on campus roads. In tests with a 2 kg lateral payload, AEPGA achieved a 67.456% reduction in power consumption compared to LQR over 22 s. For path tracking, a 3D LiDAR system was integrated. On a 77 m indoor corridor, the system reached an average speed of 1.39 km/h with a 0.153 m mean absolute error, while on a 190 m outdoor asphalt road, it reached 2.31 km/h with 0.283 m MAE. These results demonstrate the effectiveness of AEPGA in enhancing energy efficiency and adaptability, while maintaining reliable path tracking in varied environments.**

***Index Terms*—Adaptive equilibrium point gradual algorithm (AEPGA), linear-quadratic regulator (LQR), path tracking, reaction wheel, self-balancing bicycle.**

I. Introduction

TWO-wheel self-balancing vehicles have emerged as an interesting topic in robotics and control systems addressing the inherent instability of such vehicles, especially at low or zero speeds, using advanced control algorithms and hardware designs, and practical examples include the Segway [1], bicycles and motorcycles. Key challenges lie in selecting stabilization mechanisms and implementing suitable control methods.

Received 23 January 2025; revised 26 March 2025; accepted 14 May 2025. Date of publication 3 June 2025; date of current version 18 August 2025. Recommended by Technical Editor M. Chen and Senior Editor Z. Sun. This work was supported by the Ministry of Science and Technology, Taiwan under Grant NSTC 113-2221-E-002-158. *(Corresponding author: Chung-Hsien Kuo.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/TMECH.2025.3572225.

Digital Object Identifier 10.1109/TMECH.2025.3572225

There are three primary stabilization mechanisms employed in two-wheeled self-balancing vehicles: reaction wheel; heading steering; and moving mass. Each approach offers unique advantages but also suffers from distinct limitations. The reaction wheel mechanisms generate counter-torques to balance vehicle. The designs require considerations of the reaction wheel's size, mass and moment of inertia to achieve effective stabilization. A typical design is introduced by Siller-Alcalá et al. [2], and their highlighted the reaction wheel mechanism's simplicity and efficiency when optimized for energy consumption and stabilization performance. Zaborniak et al. [3] proposed a well-optimized reaction wheel to ensure efficient energy utilization while providing sufficient counter-torque to counteract tilting forces. Although reaction wheel mechanisms are suitable for maintaining balance at low or zero speeds, they introduce significant weight and high energy consumption due to the large moment of inertia required. On the other hand, the heading steering mechanisms are capable of adjusting the front wheel's steering angle to maintain balance as well as to counteract tilting forces. The mechanical design involves incorporating actuators and sensors to precisely control the steering angle based on feedback from the bicycle's tilt angle. While considering energy-efficient, the method that is proposed by Tanaka and Murakami [4] struggles under large tilt disturbances or low velocities. Seekhao et al. [5] proposed a bicycle robot using heading steering rely on generating centrifugal force by changing the heading angle, which is ineffective when the bicycle is stationary. Yi et al. [6] modified the trail length and caster angle of the steering mechanism that could enable balancing even at zero speed using its rail length and caster angle. Heading steering mechanisms are more energy-efficient and work well at higher speeds, but they are ineffective at low speeds or under large tilt disturbances due to the lack of sufficient forward momentum. Meanwhile, the moving mass mechanisms dynamically shift the center of gravity (CoG) or use a mass balancer to stabilize the system. This method is particularly effective at low or zero speeds because the steering-based method requires forward momentum to generate balancing forces. For instance, Lee and Ham [7] designed mechanisms for controlled movement of a mass to counteract the tilting motion of the bicycle. However, such systems are mechanically complex and often lack the responsiveness needed for rapid stabilization. Given these limitations, hybrid approaches that integrate multiple stabilization strategies [8], [9] have been developed to improve both stability and performance.

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Self-balancing bicycles have been controlled using various methods. PID control, known for its simplicity, has shown effectiveness in real-world applications for balancing robots [2], [10]. Linear quadratic regulator (LQR) [11], [12], offers better performance and stability by minimizing a cost function that balances state error and control effort. Meanwhile, model predictive control (MPC) [13] handles input and state constraints by predicting future system behavior, making it ideal for constrained systems. Kanjanawanishkul [14] stated that MPC continuously optimizes control inputs, making it particularly effective for systems with tight physical or safety limits. For robustness against uncertainties and disturbances, sliding mode control enforces a predefined sliding surface, ensuring system stability even under adverse conditions [15], [16], [17]. Several studies have also explored control methods to address uncertainties in bicycle dynamics. For example, Srinivas and Behera [18] tackled swing-up control for a reaction wheel pendulum, while Montoya and Gil-González [19], used Lyapunov stability theory to ensure stability under uncertain conditions. Hybrid approaches, like Tanaka and Murakami [4] integration of steering control with other mechanisms, and Zhu et al. [20] combination of conventional and reinforcement learning for path tracking, demonstrate versatility in addressing complex tasks.

Building upon these foundations, this article introduces several main contributions are as follows.

1) This article utilizes to transform an available fixed-gear bicycle into an autonomous self-balancing vehicle by adding a modular reaction wheel with an adjustable mounting setting, a front wheel's steering controller with a four-bar linkage, a motorized rear wheel and a battery that powers the system while acting as a counterweight for the reaction wheel's mass.
2) The LQR algorithm is refined by minimizing state-space model errors, enhancing stability and responsiveness. The integration of field-oriented control (FOC) for reaction wheel motor control improves torque and flux management, yielding smoother operation, faster responses, and reduced energy consumption.
3) The adaptive equilibrium point gradual algorithm (AEPGA) dynamically adjusts the equilibrium point using real-time IMU data, ensuring stability under varying conditions by managing reaction wheel's speed and preventing motor saturation as well as to result in less power consumption than LQR method.

The rest of this article is organized as follows. The mechanical and control system are introduced in Section II. Section III presents the optimal reaction wheel modeling and design. Section IV discusses the proposed LQR control algorithm and AEPGA method. The experimental setup and results of our proposed self-balancing bicycle are presented in Section V. Finally, Section VI concludes this article.

II. Mechatronics Design

A. Mechanism and Drive Design

This article involves a self-balancing bicycle with a fixed-gear design, eliminating brakes and complex gears to reduce weight and complexity. The self-balancing bicycle, as illustrated in Fig. 1, is approximately 1500 mm in length with 700×23C bike tires. It features a robust and precise parallel four-bar linkage mechanism for front wheel control. This linkage, as depicted in Fig. 2(a) and (b), is integrated with rod-end bearings and coupled with a motor for angle control using FOC. This enhances durability and alignment tolerance while providing precise movements compared to a belt-driven system. Comparing to the belt-driven system, this robust metal design offers greater robustness due to its rigid structure, which minimizes the risk of slipping, tolerates component's misalignments and enable controllable precise movements.

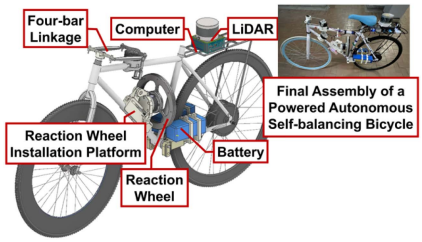


Fig. 1. Hardware architecture of the self-balancing bicycle.

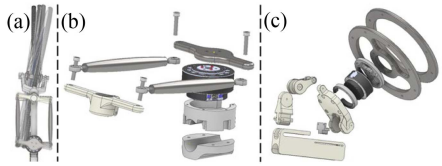


Fig. 2. Front wheel control mechanism. (a) Progressive diagram of handlebar steering mechanism, (b) Design of handlebar steering mechanism. (c) Design the reaction wheel installation platform.

A dedicated reaction wheel platform is critical for achieving effective balance control in the bicycle system. As illustrated in Fig. 2(c), the platform provides a stable mounting base that allows precise adjustments to the reaction wheel's position and angle after installation. These adjustments significantly improve the system's ability to maintain balance. During operation, the reaction wheel's torque may disperse depending on its installation angle, directly affecting its energy efficiency. Excessive dispersion can reduce overall performance, potentially compromising the kinematic model's ability to maintain balance.

To address this issue, the reaction wheel should ideally be installed perpendicular to the ground. A vertical installation ensures maximum efficiency by fully utilizing the generated torque for balance. Any deviation from vertical may cause some torque to disperse in unintended directions, reducing effectiveness, increasing energy consumption, and necessitating additional control system compensations. Furthermore, the height of the reaction wheel's installation is another critical factor. A lower CoG reduces dynamic effects, requiring less energy and minimizing torque output under disturbances. This enhances the system's ability to resist external forces and maintain stability. These design considerations, including vertical alignment and optimal placement of the reaction wheel, are crucial for improving the system's stability and control efficiency, ensuring reliable performance under various conditions.

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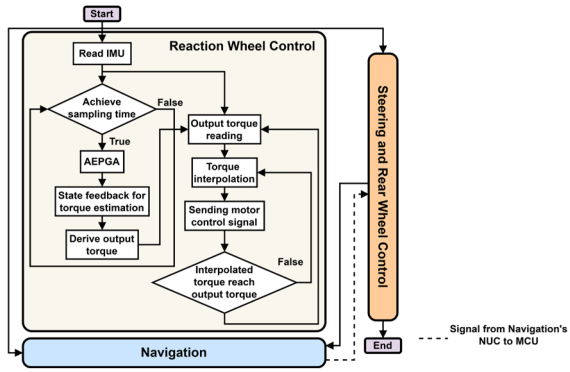


Fig. 3. Overview of control system.

The rear wheel features a hub motor controlled via PWM signals for torque adjustment, while the reaction wheel motor is managed through RS485 communication from a microcontroller. Mounted on a sliding mechanism, the reaction wheel allows fine adjustments to its angle and position relative to the frame. A 48 V battery powers motors, while a 24 V battery supplies the NUC11TNH computer, VLP-16 LiDAR, microcontrollers, and various sensors.

B. Control System Design

As depicted in Fig. 3, the control system comprises three main components: balancing control system (reaction wheel control); navigation system; and riding system (including front handle steering and rear wheel control). The system starts by activating the inertial measurement unit (IMU) and receiving command signals. During this process, a full-state feedback control and adaptive control are employed to maintain the balance. Control signals are interpolated for smoothing motor responses, reducing vibration caused by excessive signal magnitude or low frequency. The decoupled reaction wheel control and steering control systems operate independently. To ensure precise path tracking, this system utilize the pure pursuit algorithm [21] to simplify code implementation. It also integrates a VLP-16 LiDAR with FAST-LIO2 algorithm [22] for map creation and bicycle positioning. The robot operating system assists communication, while the pure pursuit algorithm calculates steering angles and rear wheel speeds, transmitting these commands to the microcontrol unit (MCU) for the control of all actuators deployed in this article.

III. Reaction Wheel Modeling and Design

A. Simplified Modeling of Reaction Wheel System

This article aims to validate the ideas of this article. To simplify the complex dynamics of a self-balancing bicycle, the bike-balancing system is modeled as an unfixed testing platform rather than being fixed to the rotational axis to perform more realistic situation with real bicycles. The simplified system is represented as a reaction wheel inverted pendulum model, illustrated in Fig. 4(a). This simplified model is constructed with 3D printed PLA material. The setup includes a brushless motor with FOC to drive the reaction wheel. Adjustable screws

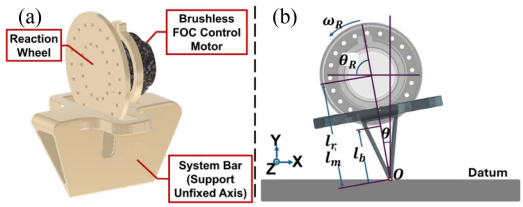


Fig. 4. (a) Simplified reaction wheel inverted pendulum model. (b) Its key physical parameters.

and nuts on the wheel plate are desired to adjust the load and moment of inertia, enhancing testing flexibility. The system's governing equations are derived using the Lagrangian equation and Newtonian kinematics.

The Lagrangian equation, based on the principle of least action, minimizes the action integral to describe system dynamics. It involves finding the Lagrangian L, a function that reaches an extremum over possible trajectories. In Lagrangian mechanics, generalized coordinates define a system's degrees of freedom, such as pendulum angle. The Lagrangian L, depending on generalized coordinates q, their time derivatives $\dot{q}$, and time t, is expressed as

$$L\ (q, \dot{q}, t) = T - V \tag{1}$$

where T and V are the system's kinetic and the potential energies, respectively. For a specific motion path, an action S can be expressed as the integral of L from time t_1 to t_2

$$S = \int_{t1}^{t2} L\ (q, \dot{q}, t)\, dt. \tag{2}$$

Then, Lagrange equations, derived from variations of L, yield n equations for a system with n degrees of freedom. A generalized form of Lagrange can be expressed as

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i\ ; L = T - V \tag{3}$$

where $\dot{q}_i$ and $Q_i$are generalized coordinates and corresponding forces, D is the friction dissipation.

For the reaction wheel inverted pendulum system, key physical parameters include the reaction wheel's angular velocity $\omega_R = \dot{\theta}_R$, the system's tilt angle θ, the distance from the motor's CoG to the pivot l_m, the distance from the reaction wheel's CoG to the pivot l_r. These parameters, as illustrated in Fig. 4(b), define the system's dynamics. Once the physical parameters are defined, the system's governing equations can be derived using the Lagrangian method. The frictional dissipation generated by the relative motion of the contact surfaces can be expressed as

$$D = \frac{1}{2} b_b \dot{\theta}^2 \tag{4}$$

where b_b is the ground friction. Since relative motion at the contact surface is negligible, friction is assumed to be nearly zero. The total kinetic energy of the system is calculated based on the rotational motion of its components ω. By fixing the reference frame at the center of mass of the object, the relationship between rotational kinetic energy $E_{k,\text{rotation}}$ and the moment of inertia of

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the object I can be expressed as

$$E_{k,\text{rotation}} = \frac{1}{2} I\omega^2 \tag{5}$$

where I includes contributions from the reaction wheel's moment of inertia with its center I_R, the motor's moment of inertia with the pivot I_{mo}, the bracket's moment of inertia with the pivot I_{bo} and the rotation of reaction wheel's moment of inertia with the pivot I_{ro}. The total kinetic energy can be calculated as

$$T = \frac{1}{2} I_{bo}\dot{\theta}^2 + \frac{1}{2} I_{mo}\dot{\theta}^2 + \frac{1}{2} I_{ro}\dot{\theta}^2 + \frac{1}{2} I_R\dot{\theta}_R^2. \tag{6}$$

The potential energy V, based on gravitational forces, is

$$V = g\left(M_b l_b + M_r l_r + M_m l_m\right)\cos\theta \tag{7}$$

where M_b, M_r, and M_m are the masses of the pendulum frame, reaction wheel and the motor, respectively; g is the gravity acceleration; l_b, l_r, and l_m can be found in Fig. 4. Using generalized coordinates as mentioned before , the Lagrange equations can be obtained as

$$I_{bo}\ddot{\theta} + I_{mo}\ddot{\theta} + I_{ro}\ddot{\theta} - g\left(M_b l_b + M_r l_r + M_m l_m\right)\sin\theta + b_b\dot{\theta} = 0 \tag{8}$$

$$I_R\,\ddot{\theta}_R = 0. \tag{9}$$

Combining (8) and (9), a nonlinear, time-invariant governing equation is obtained as

$$\ddot{\theta} + \frac{b_b}{I_{so}}\dot{\theta} - \frac{g\left(M_b l_b + M_r l_r + M_m l_m\right)}{I_{so}}\sin\theta = -\frac{I_R}{I_{so}}\ddot{\theta}_R. \tag{10}$$

This equation comprehensively describes system dynamics, revealing interactions between components and forming a foundation for stability and response analysis.

B. Linearization Process for Reaction Wheel Inverted Pendulum System

Linearization is a crucial step when analyzing nonlinear systems, particularly in designing linear time-invariant controllers like the LQR. It simplifies complex dynamics by approximating system behavior near a specific operating point. Nonlinear systems often pose challenges in direct analysis and control due to their intricate behavior. Linearization, by focusing on equilibrium points, reduces this complexity. In this article, the process consists of three main steps:

1) Operating Point Calculation: The operating point is chosen by determining the equilibrium point. The equilibrium point, a stable and frequently occurring state, is identified where all state derivatives are constant, leading to

$$-g\left(M_b l_b + M_r l_r + M_m l_m\right)\sin\theta_e = 0 \tag{11}$$

where θ_e is the angle at equilibrium point. This results in two equilibrium points $\theta_e = 0$ or π. For the inverted pendulum system, the point of interest is $\theta_e = 0$.

2) Linear Approximation Using Taylor Expansion: A first-order Taylor expansion is performed around the equilibrium point $\theta_e = 0$. The states and derivatives are expanded as:

$$\theta = \theta_e + \delta\theta;\ \dot{\theta} = \dot{\theta}_e + \delta\dot{\theta};\ \ddot{\theta} = \ddot{\theta}_e + \delta\ddot{\theta}. \tag{12}$$

Substituting into the governing equations and neglecting higher order terms yields the linearized dynamic equation, as expressed as

$$\delta\,\ddot{\theta}_R = \frac{g\left(M_b l_b + M_r l_r + M_m l_m\right)}{I_r}\delta\theta - \frac{b_b}{I_r}\delta\dot{\theta} - \frac{I_{so}}{I_r}\delta\ddot{\theta}. \tag{13}$$

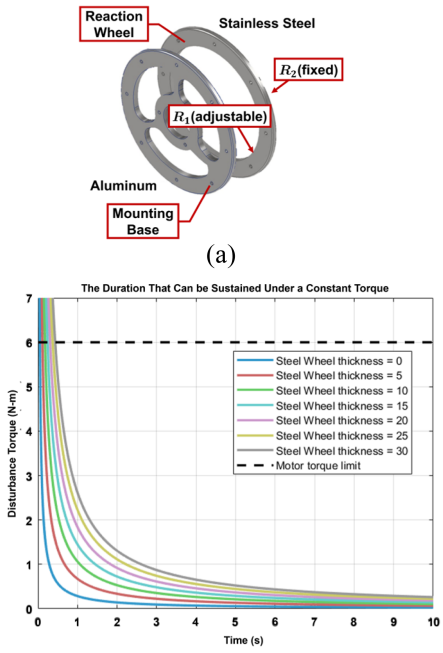


Fig. 5. (a) Reaction wheel design. (b) Time to resist constant disturbance torque relative to reaction wheel's inertia.

C. Reaction Wheel Design

Based on mathematical simplified reaction wheel model discussion in Section III-A, the reaction wheel must generate angular acceleration to produce a torque $\tau_R$counteracting the torque caused by a tilt angle θ. The reaction wheel module of the propose self-balancing bicycle is designed with an aluminum base and a high-density stainless-steel layer, as shown in Fig. 5(a). Such a configuration is to flexibly adjust the overall moment of inertia by additionally setting the moment of inertia of the high-density stainless-steel hollow cylindrical wheel disk. It is noted that the aluminum base has a moment of inertia of 0.00841681 N·m^2.

The high-density stainless-steel hollow cylindrical wheel disk is designed with a fixed mass m with thickness h; the outer radius R_2; the inner radius R_1 . For a fixed torque generated by the reaction wheel, a smaller moment of disk inertia I_R allows the reaction wheel to achieve maximum speed faster, and it highlights the inverse relationship between I_R and angular acceleration $\ddot{\theta}_R$. When the reaction wheel counteracts a tilt angle θ, it must generate an angular acceleration to produce a torque τ_R that opposes the torque caused by the tilt. It is noted that the

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motor used in this article has a maximum angular velocity of 33.93 rad/s.

The design goal is to increase the moment of inertia and improve stability under disturbances. The moment of inertia I_R, related to R_1 and R_2, is expressed as

$$I_R = \frac{1}{2} m \left(R_1^2 + R_2^2\right). \tag{14}$$

Increasing R_1 leads to increase I_R, but with a constant m, increasing R_1 also affects the material density ρ

$$m = \pi \rho h \left(R_2^2 - R_1^2\right) \tag{15}$$

where h is the reaction wheel's thickness.

To meet these requirements, the motor, rated at 6 N·m torque, relies on the reaction wheel to resist disturbances until the motor reaches its speed limit ω of 33.93 rad/s. Hence, with $\omega_0 = 0$, the time to the saturation which determine disturbance resistance, is governed by

$$\tau = I \frac{\omega - w_0}{t}. \tag{16}$$

The time to resist constant disturbance torque relative to reaction wheel's inertia w.r.t. different thicknesses of the stainless-steel hollow cylindrical wheel disk is shown in Fig. 5(b). According to the mass limitation of the stainless-steel hollow cylindrical wheel disk and the applied motor's torque specification (i.e., 6 N·m), the thickness of the stainless-steel layer is selected as 10 mm, and it contributes an additional 0.022897396 N·m^2. It is noted that the thickness equaling to 0 means that only the aluminum base is used.

IV. Optimal Control Algorithm for Reaction Wheel Linear Time-Invariant System

A. Applying Conventional LQR as a Beginning

This article initially employs the LQR [12] due to its simple mathematical framework, optimal performance guarantees, and the flexibility to adjust system behavior using user-defined weighting matrices. The linearized motion equation of the reaction wheel system is expressed as

$$\frac{g\left(M_b l_b + M_r l_r + M_m l_m\right)}{I_r}\theta - \frac{b_b}{I_r}\dot{\theta} - \frac{I_{so}}{I_r}\ddot{\theta} - \ddot{\theta}_R = 0. \tag{17}$$

Since direct measurement of the reaction wheel velocity $\dot{\theta}$ is challenging, expressing relative to the pendulum's motion as

$$\dot{\theta}_R = \omega_{R,\text{self}} + \dot{\theta} \tag{18}$$

where $\omega_{R,\text{self}}$ the angular velocity of the reaction wheel relative to its axis. Defining state variables as $x_1 = \theta$; $x_2 = \dot{\theta}$; $x_3 = \omega_{R,\text{self}}$, the state-space representation of the system is then derived as

$$\begin{bmatrix} \dot{x_1} \\ \dot{x_2} \\ \dot{x_3} \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ A & -B & \frac{b_R}{I_{so}} \\ -A & B & -b_R C \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{k_t}{I_{so}} \\ k_t C \end{bmatrix} u(t) \tag{19}$$

where the terms of $A = \frac{g(M_b l_b + M_r l_r + M_m l_m)}{I_{so}}$, $B = \frac{b_b}{I_{so}}$, and $C = \frac{I_r + I_{so}}{I_r I_{so}}$; b_R for motor friction value for the reaction wheel motor, k_t is the motor torque constant, and $u(t)$ is the control output.

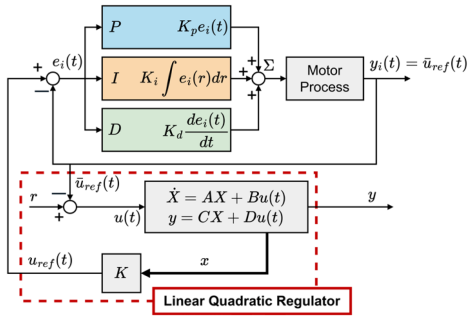


Fig. 6. Proposed optimal controller.

The LQR algorithm is then integrated into a motor PID controller, as shown in Fig. 6. The LQR controller is designed based on a linearized state-space model of the inverted pendulum system with a reaction wheel. The objective is to minimize a quadratic cost function $\int (x^T Q x + u^T R u) \mathrm{d}t$, where x is the system states and u is the control input (motor current). The Q and R matrices were tuned experimentally to balance tracking accuracy and control effort. The resulting gain matrix K was obtained by solving the Riccati equation and applied via full-state feedback.

B. Adaptive Equilibrium Point Gradual Algorithm (AEPGA)

LQR alone assumes a fixed equilibrium point, typically at zero tilt, based on a linearized model around that point. Under normal condition without external loads, the IMU outputs remain near zero, indicating a stable system. However, when the CoG shifts due to added payloads or asymmetrical loading, the actual balance point of the system changes to an unknown angle θ_p. In such cases, the LQR controller continues to regulate the system toward incorrect reference, leading to steady-state errors, increased motor effort, or even failure to balance. Therefore, an adaptive mechanism is needed to track and update the true equilibrium point in real-time.

Addressing this issue, an adaptive algorithm, named AEPGA was implemented in the MCU gradually to refine the balance point. The AEPGA dynamically estimates and updates the system's true equilibrium point using real-time IMU feedback. Specifically, the algorithm maintains a sliding average of recent IMU readings to track changes in the steady-state tilt angle. It determines the difference from the prior balance point, scales the difference by a weight K_s, and updates the balance point. This iterative process, expressed as:

$$x_i^{k+1} = x_{i+1}^k \tag{20}$$

$$E_R = \frac{1}{n} \sum_{i=1}^{n} x_i^{k+1} \tag{21}$$

$$E_e^{k+1} = E_e^k + \left(E_R - E_e^k\right) K_s; 0 < K_S < 1 \tag{22}$$

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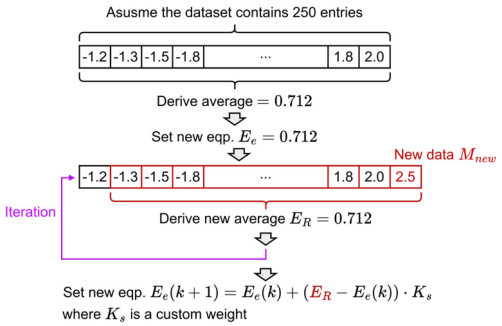


Fig. 7. Example of the proposed adaptive equilibrium point gradual algorithm (AEPGA).

where x_i is ith IMU reading with x_1^k as the latest value, E_R is the average of x^{k+1}, E_e^k is previous estimated balanced point and E_e^{k+1} is the updated balanced point, and K_S is a small positive gain that controls the convergence speeds. To speed up the convergence of the adaptive point, AEPGA favors shorter-term IMU data, though this can cause more fluctuations. The gain K_S, helps dampen these effects. However, higher K_S values increase overshoot in the adaptive angle response.

The estimated equilibrium point E_e^k is gradually adjusted using the update rule as (22). This updated value E_e^{k+1} is then treated as the new reference for the balance controller, effectively shifting the control target to match the real CoG. This enables the system to compensate for asymmetric loading conditions without saturating the motor or introducing steady-state bias. This enables the system to compensate for asymmetric loading conditions without saturating the motor or introducing steady-state bias. An example of the AEPGA is illustrated in Fig. 7.

V. Experimental Results and Analysis

A. Simulation and Validation of the Optimized Control Model for the Simplified Reaction Wheel System

This section validates the accuracy and effectiveness of the optimized control model for the reaction wheel system by comparing simulated and experimental pulse responses. Using MATLAB Simulink, the LQR controller was employed to simulate the system's behavior, which was then evaluated against a physical reaction wheel system. Key parameters, including moments of inertia, friction coefficients, and motor torque constants, were calibrated. Friction was approximated as negligible, and unit conversions were performed to align motor outputs (in degrees) with the model (in radians). A conversion gain of $\vec{K}_{\text{real}} = 0.91675\vec{K}$ was applied to match the microcontroller's weights. To verify the proposed linearization to the dynamic model, a MATLAB simulation experiment was conducted on the small platform of the reaction wheel. When the system deviates from the equilibrium point by approximately 1.5 rads, the model starts to become inaccurate, as shown in Fig. 8.

Moreover, the pulse response experiment evaluates how accurately the linearized reaction wheel control model imitates real-world behaviors. By applying the identical pulse inputs at 1.6 A, 1.24 A, and 0.98 A to both the simulation and the

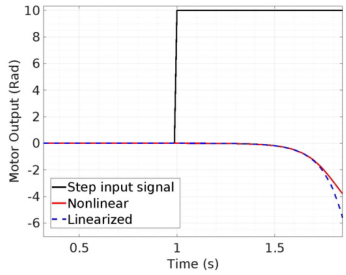


Fig. 8. Comparison of proposed linearized and nonlinear systems on the small reaction wheel platform.

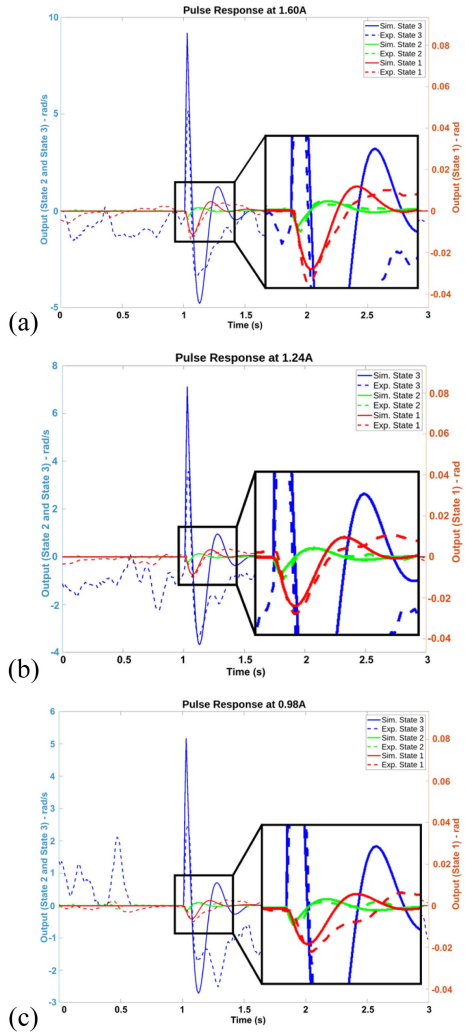


Fig. 9. Comparison of simulated (sim.) and experimental (exp.) platform pulse responses at different input-current amplitudes. (a) 1.6 A. (b) 1.24 A. (c) 0.98 A.

actual system, three principal states are observed: the pendulum angle θ (state 1), the pendulum angular velocity $\dot{\theta}$ (state 2), and the reaction wheel's angular velocity $\omega_{R,\text{self}}$ (state 3). As illustrated in Fig. 9, the first two states θ and $\dot{\theta}$ (red and green lines) exhibit a close agreement between simulated (Sim.) and experimental (Exp.) responses, confirming the model's validity for these parameters. However, the third state $\omega_{R,\text{self}}$ shows more variation near peak speeds, suggesting limits in the motor's

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TABLE I
MAE AND RMSE ANALYSIS OF THE SYSTEM'S STATES BETWEEN SIMULATED AND EXPERIMENTAL PULSE RESPONSES

Pulse	State 1		State 2		State 3	
Response	MAE	RMSE	MAE	RMSE	MAE	RMSE
1.60 A	0.0014	0.0019	0.0246	0.0353	0.7919	1.0280
1.24 A	0.0016	0.0020	0.0187	0.0270	0.9069	1.0632
0.89 A	0.0024	0.0028	0.0227	0.0305	0.8669	1.0959

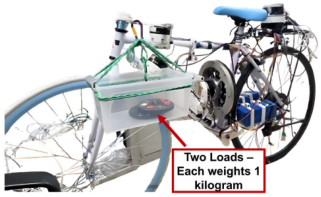


Fig. 10. Experiment setup of adaptive balance point for bicycle reaction wheel system.

observer precision and potential unmodeled nonlinearities. Nevertheless, the matching slopes of the response curves (black box) imply that the simulated torque closely matches the actual torque produced by the motor. Furthermore, smaller input amplitudes, i.e., 0.98 A, achieve quicker stabilization and weaker oscillations, demonstrating the controller's adaptability under various input conditions.

Table I presents mean absolute error (MAE) and root mean square error (RMSE) for each pulse amplitude and state. Notably, state 1 and state 2 have lower errors compared to state 3, providing the observed discrepancies in the reaction wheel's speed. Although the state 3 shows larger differences between simulation and experiment, this does not compromise the overall reliability of the linearized model for other states.

B. Adaptive Balance Point Experiment for Bicycle Reaction Wheel System

This experiment utilizes weighted plates to simulate changes in the bicycle's CoG, as shown in Fig. 10. Each plate weighs 1 kg. The experiment involves sequentially adding two plates and then removing them to observe state variations and compare systems with and without the adaptive algorithm. During data collection of reaction wheel for analysis, any external force interference may affect the comparability of the experiment. Therefore, all the data of the reaction wheel was collected when the bicycle's speed was zero.

The proposed system integrated a full-state feedback controller with the adaptive AEPGA as shown in Fig. 11(a). AEPGA adaptively updates the balance point based on tilt angle changes, enabling quick respond to CoG shifts. This reduces oscillation and improve energy efficiency, achieving a 67.456% reduction in. power compared to the standard LQR over a 22-s test (the area under the blue curve as total angles for energy consumption comparison bases). Without AEPGA as shown in Fig. 11(b), CoG shifts cause the reaction wheel to oscillate and exceed safe speed limit (∼35 rad/s), leading to instability and failure. Furthermore, AEPGA maintains balance even under large CoG changes, such as during turns or slope riding. As shown in

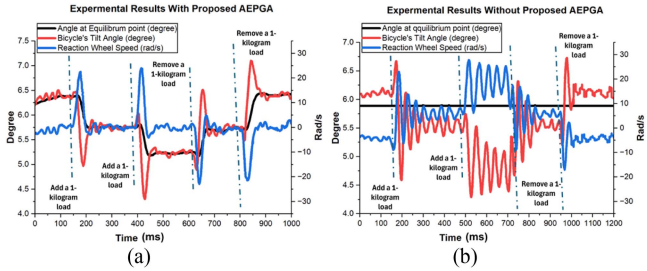


Fig. 11. Experiment results (a) with and (b) without the proposed AEPGA.

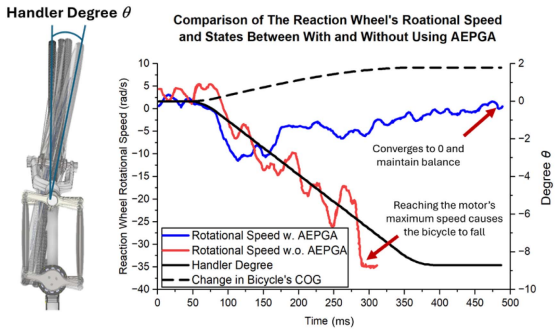


Fig. 12. Comparison of the reaction wheel speed and states between with (blue) and without (red) using the AEPGA when the bicycle is turning.

Fig. 12, the system fails without AEPGA at a steering angle of -6.3° and CoG shift of 1.6°, while with AEPGA, all states remain stable, demonstrating its effectiveness under dynamic conditions.

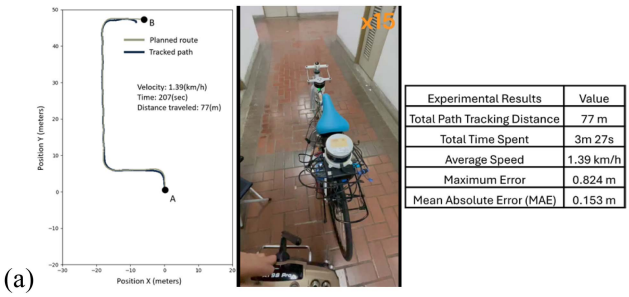
C. Comparative Evaluation With Existing Controllers

Table II gives a qualitative comparison of AEPGA with other existing controllers based on theoretical features and reported characteristics in literature. Quantitative comparisons are not included due to lack of uniform experimental conditions among the methods.

The AEPGA controller outperforms traditional methods like PID and LQR [12] by adaptively adjusting the balance point in real time, maintaining stability under CoG shifts, steering, and slope riding. It also achieves a 67% reduction in energy compared to LQR. Unlike sliding mode-based methods, e.g., ELM+ITSM [23], AEPGA avoids chattering and complex tuning, while remaining lightweight and MCU-friendly. Although ELM-based controllers offer strong robustness, they require higher computational effort.

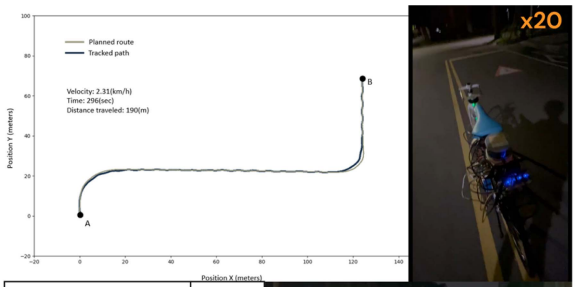
Compared to IDA-PBC+FL [24], AEPGA offers a simpler, more implementation-friendly solution without sacrificing stability. While IDA-PBC+FL provides theoretical global stability, it requires complex nonlinear modeling and controller switching. In contrast, AEPGA achieves practical stability with real-time adaptability to CoG shifts and disturbances, significantly reducing energy consumption and supporting autonomous navigation.

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Experimental Results	Value
Total Path Tracking Distance	77 m
Total Time Spent	3m 27s
Average Speed	1.39 km/h
Maximum Error	0.824 m
Mean Absolute Error (MAE)	0.153 m

(a)



Experimental Results	Value
Total Path Tracking Distance	190 m
Total Time Spent	4m 56s
Average Speed	2.31 km/h
Maximum Error	1.44 m
Mean Absolute Error (MAE)	0.283 m

(b)

Fig. 13. (a) Indoor corridor path tracking. (b) Outdoor path tracking experiment and maximum self-balancing speed test.

TABLE II
COMPARISON WITH EXISTING CONTROLLERS

Feature/ Method	AEPGA (Ours)	PID	LQR	EML + ITSM [23]	IDA-PBC + FL [24]
Adaptivity	✓ (adaptive balance point)	× (fixed gain)	× (fixed model gain)	× (learn uncertainty via ELM)	×
Disturbance Rejection	✓ (handle unknown payload)	× (weak under disturbance)	× (moderate)	✓	✓ (theorical prove)
Stability Guarantee	✓ (adaptive drift correction)	× (local)	× (local)	✓ (finite-time - Lyapunov)	✓ (Global - Lyapunov)
Chattering	✓	×	×	✓ (need reduce with boundary layer)	×
Energy Efficiency	✓ (moderate)	× (inefficient under dynamic load)	× (higher torque usage)	-	-
Complexity	✓ (moderate)	✓ (simple)	✓ (moderate)	× (high)	× (high)
Robustness to CoG Shift	✓ (real time IMU-based)	×	×	✓ (moderate)	×
Navigation Support	✓	×	×	×	×

D. Path Tracking Experiment of the Self-Balancing Bicycle Reaction Wheel System

This article conducted path tracking tests for the self- balancing bicycle system in both indoor and outdoor environments. The experiments were carried out at the College of Engineering Building at National Taiwan University (NTU). For the indoor experiment, as depicted in Fig. 13(a), the bicycle navigated a path of 77 m in 3 min and 27 s at an average speed of 1.39 km/h. The system exhibited precise path tracking with a maximum error of 0.824 m and a MAE of 0.153 m. However, narrow corridors and tight corners limited speed, as the pure pursuit algorithm struggled with sharp turns.

In contrast, the outdoor experiment, as depicted in Fig. 13(b), covered a longer path of 190 m in 4 min and 56 s at an average speed of 2.31 km/h. The system faced challenges such as uneven terrain, slopes, and wind, resulting in a higher maximum error of 1.44 m and a MAE of 0.283 m. Despite these disturbances, the full-state feedback control and adaptive balance point algorithm ensured successful path tracking. The experiments highlighted the need for improved cornering and speed adjustment for indoor environments and enhanced disturbance rejection for outdoor scenarios. The maximum speed of 17 km/h for the self-balancing test was also validated in NTU campus.

VI. CONCLUSION

This article developed an autonomous self-balancing bicycle system capable of maintaining stability by utilizing a reaction wheel to stabilize gravitational torque during body tilt. A novel AEPGA adaptive balance point control mechanism dynamically aligned the balance equilibrium point, enhancing resistance to external changes such as shifts in the CoG and uneven terrains. The imbalanced lateral payload experiments were for power consumption validation. The results showed that the proposed AEPGA outperformed the LQR with 67.456% total power efficiency reduction during 22 s tests to show the advantage on power consumption on the reaction wheel. While 3D LiDAR technology enabled path tracking experiments in indoor and outdoor path tracking scenarios, and experimental results demonstrated promising outcomes in under turnings, uneven terrains. Moreover, high speed riding was also arranged, and a maximum speed of 17 km/h was achieved.

However, current limitations in carrying significant payloads due to motor and space constraints result in the needs for a coupled system design that integrates balance control with real-time CoG adjustments. Future work will focus on enabling high-speed operation, addressing centrifugal force-induced challenges, and exploring broader applications of the reaction wheel model in bionic and mobile robots. These efforts aim to enhance the system's adaptability, stability, and performance, contributing to the advancement of lightweight, high efficiency autonomous vehicles and robotics.

REFERENCES

[1] R. Ďuriš, E. Labašová, J. Oravcová, and Š. Šimon, "A two wheel self-balancing vehicle," *Res. Papers Fac. Mater. Sci. Technol. Slovak Univ. Technol.*, vol. 29, no. 49, pp. 78–93, Sep. 2021, doi: 10.2478/rput-2021-0027.

[2] I. I. Siller-Alcalá, J. U. Liceaga-Castro, R. A. Alcántara-Ramírez, and S. Calzadilla-Ayala, "Using a flywheel to stabilize a self-balancing bicycle," *WSEAS Trans. Syst. Control*, vol. 19, pp. 73–84, Apr. 2024, doi: 10.37394/23203.2024.19.8.

Authorized licensed use limited to: The University of Hong Kong Libraries. Downloaded on August 16,2026 at 03:51:30 UTC from IEEE Xplore. Restrictions apply.

[3] D. Zaborniak, K. Patan, and M. Witczak, "Design, implementation, and control of a wheel-based inverted pendulum," *Electronics*, vol. 13, no. 3, Jan. 2024, Art. no. 514, doi: 10.3390/electronics13030514.

[4] Y. Tanaka and T. Murakami, "Self sustaining bicycle robot with steering controller," in *Proc. 8th IEEE Int. Workshop Adv. Motion Control*, Mar. 2004, pp. 193–197, doi: 10.1109/AMC.2004.1297665.

[5] P. Seekhao, K. Tungpimolrut, and M. Parnichkun, "Development and control of a bicycle robot based on steering and pendulum balancing," *Mechatronics*, vol. 69, Aug. 2020, Art. no. 102386, doi: 10.1016/j.mechatronics.2020.102386.

[6] J. Yi, D. Song, A. Levandowski, and S. Jayasuriya, "Trajectory tracking and balance stabilization control of autonomous motorcycles," in *Proc. IEEE Int. Conf. Robot. Automat.*, 006, pp. 2583–2589, doi: 10.1109/ROBOT.2006.1642091.

[7] S. Lee and W. Ham, "Self stabilizing strategy in tracking control of unmanned electric bicycle with mass balance," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2002, pp. 2200–2205, vol. 3, doi: 10.1109/IRDS.2002.1041594.

[8] L. Keo and M. Yamakita, "Control of an autonomous electric bicycle with both steering and balancer controls," *Adv. Robot.*, vol. 25, no. 1–2, pp. 1–22, Jan. 2011, doi: 10.1163/016918610X538462.

[9] P. Wang, Y. Gong, J. Yi, and T. Liu, "An integrated stationary/moving balance control of an autonomous bikebot," in *Proc. Amer. Control Conf.*, Jul. 2019, pp. 3273–3278, doi: 10.23919/ACC.2019.8814916.

[10] C. -H. Chiu and C. -Y. Wu, "Bicycle robot balance control based on a robust intelligent controller," *IEEE Access*, vol. 8, pp. 84837–84849, 2020, doi: 10.1109/ACCESS.2020.2992792.

[11] Z. Yongli, Y. Liu, and G. Yi, "Model analysis of unmanned bicycle and variable gain LQR control," in *Proc. IEEE 9th Joint Int. Inf. Technol. Artif. Intell. Conf.*, Dec. 2020, pp. 1196–1200, doi: 10.1109/ITAIC49862.2020.9338891.

[12] V. L. Mehrmann, Ed., "The autonomous linear quadratic control problem," in *Lecture Notes in Control and Information Sciences*, vol. 163. Berlin, Germany: Springer, 1991, doi: 10.1007/BFb0039443.

[13] C. E. García, D. M. Prett, and M. Morari, "Model predictive control: Theory and practice—A survey," *Automatica*, vol. 25, no. 3, pp. 335–348, May 1989, doi: 10.1016/0005-1098(89)90002-2.

[14] K. Kanjanawanishkul, "LQR and MPC controller design and comparison for a stationary self-balancing bicycle robot with a reaction wheel," *Kybernetika*, vol. 51, pp. 173–191, Mar. 2015, doi: 10.14736/kyb-2015-1-0173.

[15] D. Gutiérrez-Oribio, Á. Mercado-Uribe, J. A. Moreno, and L. Fridman, "Robust global stabilization of the third order reaction wheel pendulum system∗," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 5111–5116, Jan. 2020, doi: 10.1016/j.ifacol.2020.12.1147.

[16] D. Gutiérrez-Oribio, Á. Mercado-Uribe, J. A. Moreno, and L. Fridman, "Joint swing-up and stabilization of the Reaction wheel pendulum using discontinuous integral algorithm," *Nonlinear Anal. Hybrid Syst.*, vol. 41, Aug. 2021, Art. no. 101042, doi: 10.1016/j.nahs.2021.101042.

[17] M. Defoort and T. Murakami, "Sliding-mode control scheme for an intelligent bicycle," *IEEE Trans. Ind. Electron.*, vol. 56, no. 9, pp. 3357–3368, Sep. 2009, doi: 10.1109/TIE.2009.2017096.

[18] K. N. Srinivas and L. Behera, "Swing-up control strategies for a reaction wheel pendulum," *Int. J. Syst. Sci.*, vol. 39, no. 12, pp. 1165–1177, Dec. 2008, doi: 10.1080/00207720802095137.

[19] O. D. Montoya and W. Gil-González, "Nonlinear analysis and control of a reaction wheel pendulum: Lyapunov-based approach," *Eng. Sci. Technol. Int. J.*, vol. 23, no. 1, pp. 21–29, Feb. 2020, doi: 10.1016/j.jestch.2019.03.004.

[20] X. Zhu et al., "Online series-parallel reinforcement-learning- based balancing control for reaction wheel bicycle robots on a curved pavement," *IEEE Access*, vol. 11, pp. 66756–66766, 2023, doi: 10.1109/ACCESS.2023.3268524.

[21] P. T.-T. Nguyen, S. -W. Yan, J. -F. Liao, and C. -H. Kuo, "Autonomous mobile robot navigation in sparse LiDAR feature environments," *Appl. Sci.*, vol. 11, no. 13, Jun. 2021, Art. no. 5963, doi: 10.3390/app11135963.

[22] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast direct LiDAR-inertial odometry," *IEEE Trans. Robot.*, vol. 38, no. 4, pp. 2053–2073, Aug. 2022, doi: 10.1109/TRO.2022.3141876.

[23] L. Chen, B. Yan, H. Wang, K. Shao, E. Kurniawan, and G. Wang, "Extreme-learning-machine-based robust integral terminal sliding mode control of bicycle robot," *Control Eng. Pract.*, vol. 121, Apr. 2022, Art. no. 105064, doi: 10.1016/j.conengprac.2022.105064.

[24] L. Cui et al., "Nonlinear balance control of an unmanned bicycle: Design and experiments," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Oct. 2020, pp. 7279–7284, doi: 10.1109/IROS45743.2020.9341150.



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